

Complementary events, operations on events and Euler–Venn diagrams

Events are sets; the diagram does the bookkeeping the algebra would otherwise have to.

LESSON 81–82

2 × 40 MIN

ALGEBRA 11, §30

IGX 24.3

LESSON CLOCK

15'

25'

25'

20'

5'

Events as sets	15 min
The addition rule	25 min
Filling a Venn diagram	25 min
De Morgan and three sets	20 min
Homework	5 min

LEARNING OBJECTIVES

- Form the union, intersection and complement of events.
- Read and fill an Euler–Venn diagram from given totals.
- Use the addition rule, and recognise mutually exclusive events.
- Use De Morgan’s laws to rewrite a compound event.

TEXTBOOK

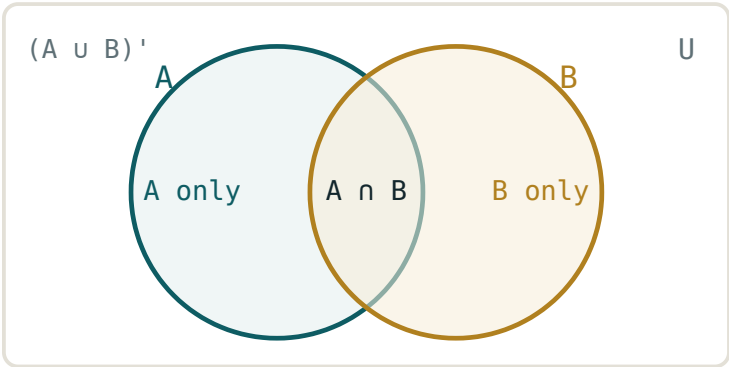
National	pp. 353–364
Cambridge	Core & Extended, pp. 635–638
Workbook	Exercise 24.3

01

Explanation

Events as sets

IN WORDS	NOTATION	PROBABILITY
A or B	$A \cup B$	$P(A) + P(B) - P(A \cap B)$
A and B	$A \cap B$	from the data, or $P(A)P(B)$ if independent
not A	A'	$1 - P(A)$
A but not B	$A \cap B'$	$P(A) - P(A \cap B)$
neither	$(A \cup B)'$	$1 - P(A \cup B)$



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Four regions, and every event in the table is one or two of them.

COMPLEMENTARY, EXCLUSIVE, EXHAUSTIVE

Two events are **complementary** when they are both exclusive ($A \cap B = \emptyset$) and exhaustive ($A \cup B = U$). Then and only then $P(B) = 1 - P(A)$. Exclusive alone is not enough.

The addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The subtraction removes the double count. For mutually exclusive events the overlap is empty and the rule shortens.

GIVEN	$P(A \cup B)$	$P(A \cap B)$
$P(A) = 0.6, P(B) = 0.3$	—	—
... exclusive	0.9	0
... independent	0.72	0.18
... with $P(A \cup B) = 0.8$	0.8	0.1

The last row is the standard examination move: the rule is used **backwards** to find the overlap from the union.

EXCLUSIVE AND INDEPENDENT ARE DIFFERENT, AND INCOMPATIBLE

Exclusive: $P(A \cap B) = 0$. Independent: $P(A \cap B) = P(A)P(B)$. If both probabilities are positive, the two cannot hold together — a question that says an event is “exclusive and independent” contains an error, or one of the probabilities is zero.

Filling a Venn diagram

Given totals, always fill the **overlap first**, then work outwards. Every region must be a count of things counted once.

Example. Of 40 pupils, 25 study physics, 18 chemistry, 10 both.

REGION	COUNT	ORDER TO FILL
$P \cap C$	10	first
P only	$25 - 10 = 15$	second
C only	$18 - 10 = 8$	third
neither	$40 - 33 = 7$	last

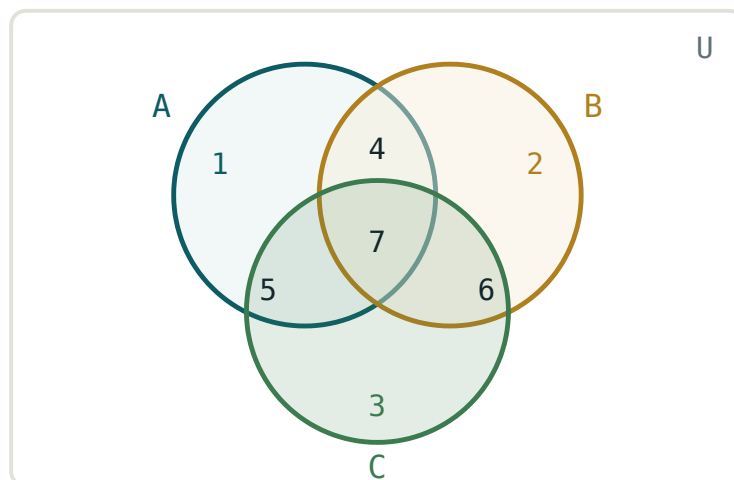
Now every probability is one division: $P(\text{neither}) = \frac{7}{40}$, $P(\text{physics only}) = \frac{15}{40} = \frac{3}{8}$, $P(P \mid C) = \frac{10}{18} = \frac{5}{9}$.

THE FOUR REGIONS MUST TOTAL $|U|$
 $15 + 10 + 8 + 7 = 40$. Do this check every time; it catches the commonest error, which is writing 25 in the “physics only” region instead of 15.

De Morgan and three sets

$$(A \cup B)' = A' \cap B' \quad (A \cap B)' = A' \cup B'$$

In words: “not (A or B)” is “not A **and** not B”. The connective flips when the bar is distributed.



seven regions inside, one outside

*Three events, seven regions inside and one outside —
the same filling method.*

With three events the addition rule grows:

$$P(A \cup B \cup C) = \Sigma P(A) - \Sigma P(A \cap B) + P(A \cap B \cap C)$$

— add the singles, subtract the pairs, add the triple back. This is inclusion–exclusion, and the alternating signs continue for any number of events.

FILL THREE SETS FROM THE MIDDLE OUT

Start with $A \cap B \cap C$, then the three pairwise regions (each pair total **minus** the middle), then the three singles, then the outside. Any other order double-counts.

02 Worked examples

EXAMPLE 1 Given $P(A) = 0.6$, $P(B) = 0.3$ and $P(A \cup B) = 0.8$, find $P(A \cap B)$ and $P(A \cap B')$.

$$① \quad 0.8 = 0.6 + 0.3 - P(A \cap B)$$

$$② \quad P(A \cap B) = 0.1$$

$$③ \quad P(A \cap B') = 0.6 - 0.1 = 0.5$$

ANSWER

0.1 and 0.5

EXAMPLE 2 Of 40 pupils, 25 study physics, 18 chemistry and 10 both. Find $P(\text{neither})$ and $P(\text{exactly one})$.

$$① \quad \text{Overlap } 10; \text{ physics only } 15; \text{ chemistry only } 8.$$

$$② \quad \text{Neither } 40 - 33 = 7.$$

$$③ \quad P(\text{neither}) = \frac{7}{40} = 0.175$$

$$④ \quad P(\text{exactly one}) = \frac{23}{40} = 0.575$$

ANSWER

0.175 and 0.575

EXAMPLE
3**Show that $(A \cup B)' = A' \cap B'$ on a Venn diagram, and use it to find $P(\text{neither})$ when $P(A \cup B) = 0.8$.**

- ① Shade outside both circles.
That region is outside A and outside B .

② $P((A \cup B)') = 1 - 0.8$

③ $= 0.2$

ANSWER

0.2

EXAMPLE
4**Of 60 students, 30 play football, 25 chess, 20 tennis; 12 football and chess, 8 chess and tennis, 10 football and tennis, 5 all three. How many play none?**

① $30 + 25 + 20 = 75$
Singles.

② $-(12 + 8 + 10) = -30$
Pairs.

③ $+ 5 = 50$
Triple back.

④ $60 - 50 = 10$

ANSWER

10 students

03 Interactive model

Draw the two circles on the board and let the class place the four numbers before any probability is computed.

Reading a Venn diagram

$P(A \cup B)$ equals:

$$P(A) + P(B)$$

$$P(A) + P(B) - P(A \cap B)$$

$$P(A)P(B)$$

$$1 - P(A)$$

$P(A')$ equals:

$$1 - P(A)$$

$$P(A)$$

$$0$$

$$1 + P(A)$$

$(A \cup B)'$ equals:

$$A' \cup B'$$

$$A' \cap B'$$

$$A \cap B$$

$$A \cup B$$

Fill a Venn diagram starting from:

the outside

the overlap

the largest region

anywhere

Exclusive means:

$$P(A \cap B) = 0$$

$$P(A \cap B) = P(A)P(B)$$

$$P(A) = P(B)$$

$$A = B$$

For three events, the triple intersection is:

subtracted

added back

ignored

doubled

Complementary events are exclusive and:

independent

exhaustive

equal

disjointly independent

The four regions of a two-set diagram total:

$$|A| + |B|$$

$$|U|$$

$$1$$

$$|A \cap B|$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Union	Birlashma	Объединение
Intersection	Kesishma	Пересечение
Complement	To'ldiruvchi	Дополнение
Euler–Venn diagram	Eyler–Venn diagrammasi	Диаграмма Эйлера–Венна
Mutually exclusive	Birgalikda bo'lmagan	Несовместные
Exhaustive	To'liq guruh	Полная группа
De Morgan's laws	De Morgan qonunlari	Законы де Моргана
Universal set	Universal to'plam	Универсальное множество
Difference of events	Hodisalar ayirmasi	Разность событий
Partition	Bo'linish	Разбиение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The addition rule subtracts the overlap because:

it is small

it would be counted twice

of De Morgan

no reason

$(A \cap B)'$ equals:

$A' \cap B'$

$A' \cup B'$

$A \cup B$

$A \cap B$

Exclusive events with positive probability are:

independent

never independent

complementary

exhaustive

A Venn diagram is filled:

left to right

from the overlap outwards

from the outside in

randomly

For three events, pairs are:

added

subtracted

ignored

squared

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $P(A')$ when $P(A) = 0.35$

ANSWER 0.65

2. $P(A \cup B)$ when exclusive, $P(A) = 0.4$, $P(B) = 0.3$

ANSWER 0.7

3. $P(A \cap B)$ when independent, $P(A) = 0.5$, $P(B) = 0.4$

ANSWER 0.2

4. Write $(A \cup B)'$ another way

ANSWER $A' \cap B'$

5. Write $(A \cap B)'$ another way

ANSWER $A' \cup B'$

6. Of 40, 25 physics, 18 chemistry, 10 both: physics only

ANSWER 15

7. Same data: neither

ANSWER 7

Medium · the standard question

7 PROBLEMS

1. $P(A \cap B)$ when $P(A) = 0.6$, $P(B) = 0.3$, $P(A \cup B) = 0.8$

ANSWER 0.1

2. Same: $P(A \cap B')$

ANSWER 0.5

3. Same: $P(\text{neither})$

ANSWER 0.2

4. Of 40 pupils as above: $P(\text{exactly one subject})$

ANSWER 0.575

5. Same: $P(P \mid C)$

ANSWER $\frac{5}{9}$

6. Are $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$ independent?

ANSWER Yes

7. Of 60 with 30 F, 25 C, 20 T, 12 FC, 8 CT, 10 FT, 5 all: how many play none?

ANSWER 10

Hard · stretch and reason

7 PROBLEMS

1. Of 60 as above: $P(\text{exactly two sports})$

ANSWER $\frac{15}{60} = \frac{1}{4}$

2. Of 60 as above: $P(\text{football only})$

ANSWER $\frac{13}{60}$

3. If $P(A) = 0.7$, $P(B) = 0.5$, find the range of $P(A \cap B)$

ANSWER $0.2 \leq P(A \cap B) \leq 0.5$

4. If A and B are independent, show A and B' are too

ANSWER $P(A \cap B') = P(A)(1 - P(B))$

5. Three events each of probability 0.5 , pairwise independent, all three independent: $P(A \cup B \cup C)$

ANSWER 0.875

6. Of 100 people, 60 read A , 50 read B , 40 read C , 25 A and B , 20 B and C , 15 A and C , 5 all: how many read none?

ANSWER 5

7. Prove $P(A \cup B) \leq P(A) + P(B)$ for any two events

ANSWER $P(A \cap B) \geq 0$

07 Homework

Homework — 5 tasks

Draw the diagram and fill all four (or eight) regions before answering anything.

- Given $P(A) = 0.55$, $P(B) = 0.4$ and $P(A \cup B) = 0.75$, find $P(A \cap B)$, $P(A \cap B')$ and $P(\text{neither})$.
- Of 50 students, 32 study Russian, 28 English and 15 both. Draw the Venn diagram and find the probability that a student studies exactly one language.

3. State both De Morgan laws and illustrate each with a shaded Venn diagram.
4. Of 80 people, 40 own a car, 35 a bicycle, 30 a motorbike; 15 car and bicycle, 12 bicycle and motorbike, 10 car and motorbike, 5 all three. How many own none?
5. Explain in two sentences why mutually exclusive events with positive probabilities cannot be independent.